Modelling and Simulation CFD Lab Report

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1 Introduction

1.1 Investigations conducted

1. Initialising and running the simulation: CFD software ANSYS Fluent was used to simulate the 2D flow of air around an aerofoil with set operating and boundary conditions. The magnitude and velocity of the air was calculated and set for both laminar and turbulent flows.

2. Comparison of theoretical predictions to experimental data: the accuracy of the model was assessed for the turbulent flow case, since the 2-equation $k - \epsilon$ model was used as an approximation.

3. Evaluating the effect of different grid densities and discretisation schemes on solution accuracy: grid densities of 40×10, 80×20, 160×40, 320×80 (laminar) and 220×100 (turbulent) were used, with converged values of coefficients of lift and drag obtained.

4. Analysis of vector, contour and pressure coefficient plots: graphs showing velocity and static pressure distributions around the aerofoil were produced to identify boundary layer behaviours at different Reynold's numbers.

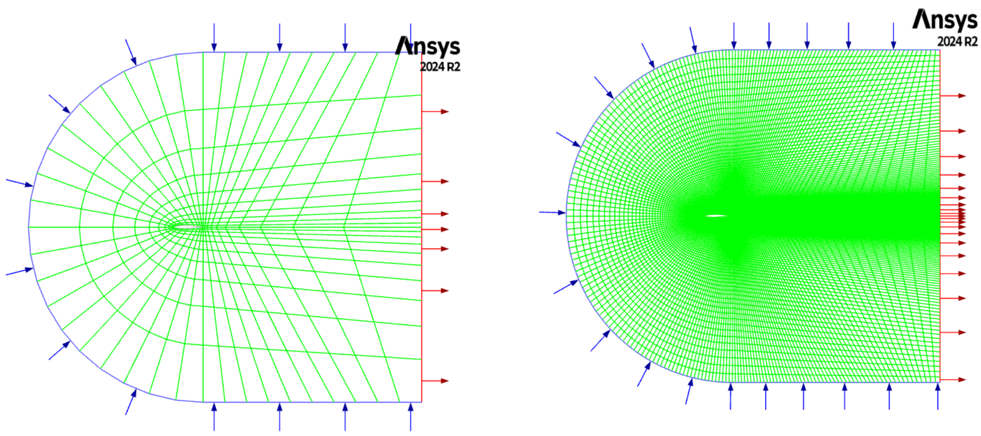

Figure 1: Aerofoil-10 (left) and Turbulent (right) C-type meshes

The grids in Figure 1, as well as the others mentioned above were used in the simulations to collect coefficient of lift (C_l) and coefficient of drag (C_d) values for both Upwind and QUICK convection schemes in order to identify the grid independent solution. For laminar flow, $\alpha = 10°$, $Re = 275$. For turbulent flow, $\alpha = 0°$, $Re = 1.2 \times 10^6$.

1.2 Report aims and objectives

- Explore the effects and accuracy of different convection schemes and grid densities.

- Make comparisons between predictions using a simple turbulence model and experimental measurements **(1)**.

2 Discussion

2.1 Vector and Contour Plots

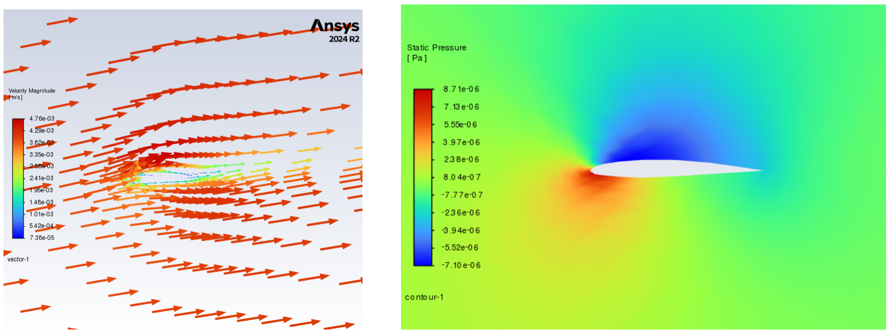

Figure 2: Aerofoil-10 Upwind vector (left) and static pressure contour (right) plots

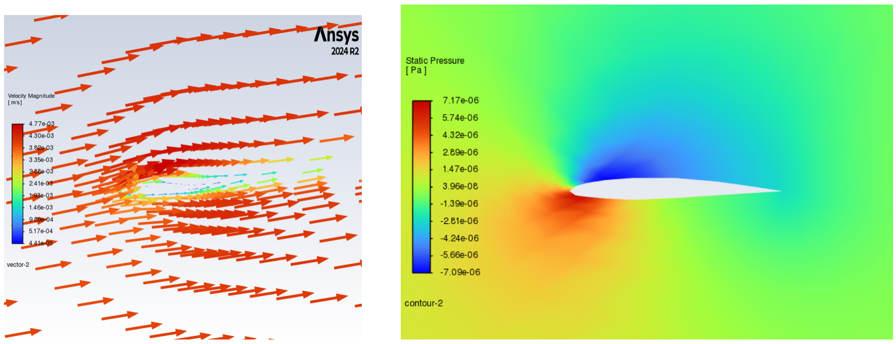

Figure 3: Aerofoil-10 QUICK vector (left) and static pressure contour (right) plots

The vector plots in Figures 2 and 3 show the velocity magnitude of the air flow around the aerofoil, however the one in Figure 2 (Upwind) produces slightly lower average velocity values because the first order discretisation scheme smooths out the values in order to eliminate sharp velocity spikes and therefore oscillations in the results. The third order QUICK scheme does the opposite, hence it is more accurate but less stable for a given grid density. Due to the higher speeds in Figure 3 (left), boundary layer separation occurs sooner, meaning that from Upwind to QUICK, C_l would be expected to decrease and C_d to increase. This is because the flow is no longer attached to the entire upper surface of the aerofoil and a wake has been created aft of the boundary layer separation.

As the grid density was increased for the laminar flow cases, it was observed that the average velocity decreased, which could be due to finer grids being able to simulate velocity gradients and viscous effects better. This would suggest that for a more refined grid, both C_l and C_d would decrease, due to a more accurate numerical simulation and less computational damping of results. The contour plots in Figures 2 and 3 reinforce the theoretical prediction of decreasing C_l and C_d values. They do this by highlighting that for the Upwind scheme, there is a higher pressure difference between the upper and lower surfaces (more blue on the upper surface) compared to the QUICK scheme, which would lead to a greater C_l for Upwind, as more lift is being produced. C_d would also be expected to decrease, since higher order discretisation schemes produce results that align closer to real-world models. For higher grid densities, a smaller pressure difference was observed, leading to the same predictions of the C_l and C_d trend as above.

2.2 Coefficient of Pressure Plots

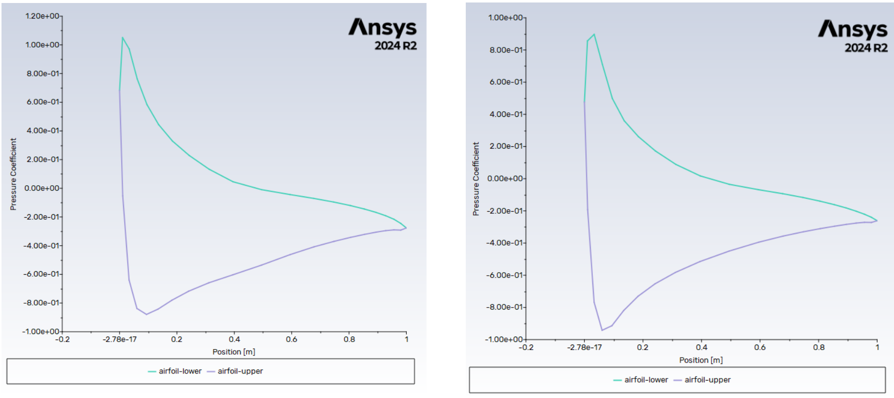

Figure 4: Aerofoil-20 Upwind (left) and QUICK (right) pressure coefficient plots

The pressure coefficient plots in Figure 4 allow one to directly infer the effects of grid density and discretisation scheme on the values of C_l and C_d. A larger pressure difference between upper and lower surfaces corresponds to a higher value of C_l and a slower upper surface pressure recovery corresponds to a higher value of C_d. As can be seen in Figure 4, a higher order discretisation scheme results in a smaller pressure difference and a faster pressure recovery, meaning a decrease in both C_l and C_d is expected. For finer grid densities, smaller pressure differences and faster pressure recovery were also observed and the opposite was for coarser grid densities.

2.3 Residual Plots

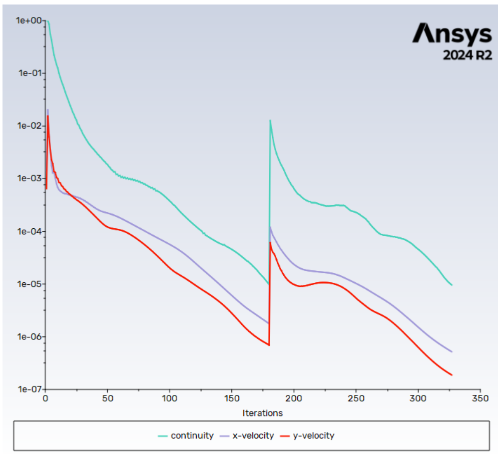

Figure 5: Aerofoil-40 Upwind followed by QUICK residual plot

Figure 5 shows that the values for C_l and C_d converge for both discretisation schemes, which was also the case for all other grid densities, proving that the results in Section 3 are reliable. The initial spike occurs due to the initial guess being far from the final solution and after several iterations, the residuals begin to converge downwards. The QUICK scheme results in a smaller final error compared to the Upwind scheme for any grid density, since it is a higher order discretisation scheme. It will therefore reach the grid independent solution as long as it has negligible error, relative to other cases.

2.4 Turbulent

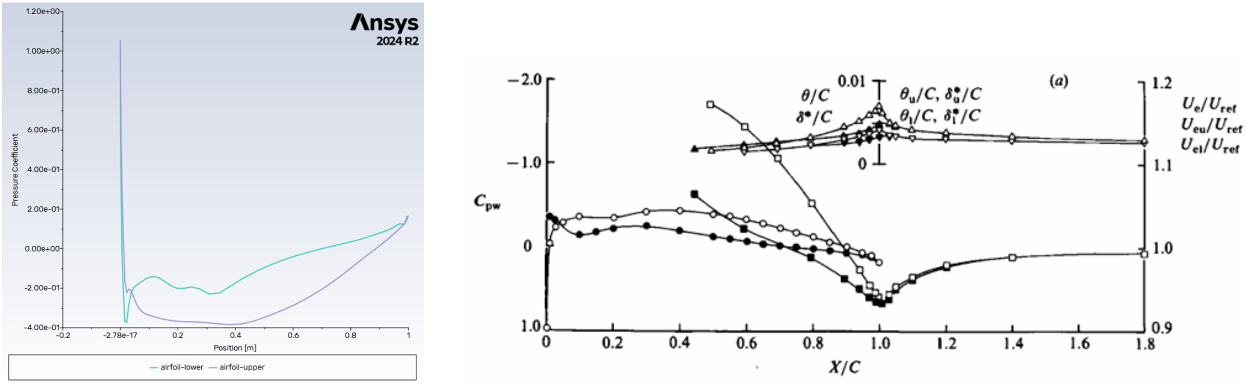

Figure 6: C_P plot of turbulent flow of ANSYS (left) and experimental data (right), Nakayama **(2)**

The pressure coefficient for both plots should be compared for chord positions of between 0.0-1.0 and for the experimental data plot, only curves with black or white circular points should be considered. When looking at this part of the graphs, it is clear that they both follow the same trend for C_P, ie. from the 0.0 to the 1.0 position, C_P increases for both upper and lower aerofoil surfaces. The value of C_P is roughly ten times smaller for the ANSYS plot compared to experimental data because the experimental data was conducted in a wind tunnel using a 2D aerofoil but in a 3D wing configuration, not a simple 2D simulation.

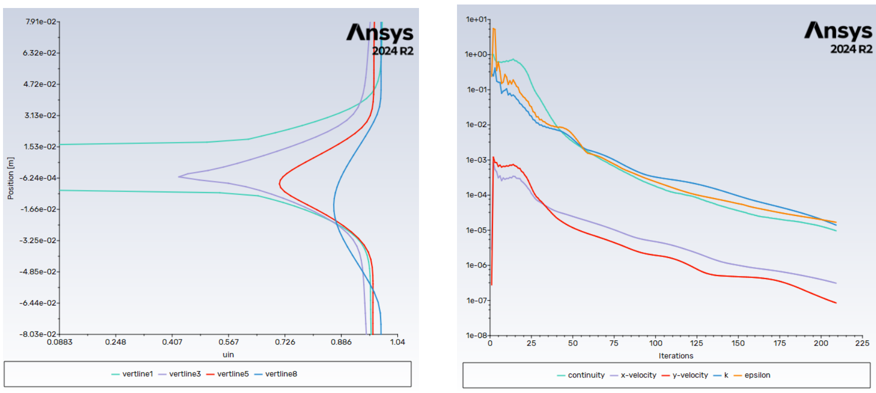

Figure 7: Turbulent aerofoil velocity profile plot (left) and turbulent residual plot (right)

x/c	Surface Label
0.893	vertline1
1.01	vertline3
1.2	vertline5
2.19	vertline8

Table 1: Streamwise positions at which experimental profiles were obtained

The velocity profile plot shows that as the airflow's streamwise position increases, the velocity difference of the air between upper and lower surfaces increases at a decreasing rate. This trend is expected of a 10%-thick conventional aerofoil, which was the model used to obtain the results in Figure 6 (right). The residual plot in Figure 7 also aligns with the experimental data in Figure 6 (black and white square points) and the experimental data provided because the C_l and C_d values converge to an accurate value, since the errors have clearly become negligible.

3 Results

3.1 $k - \epsilon$ model for turbulent flow

The 2-equation $k - \epsilon$ model and a grid with a much higher near-wall grid density were used to compute turbulent flow C_l and C_d values because they capture the steeper near-wall gradients present. In the results, only the QUICK convection of momentum scheme was used, in order to ensure accurate numerical results. A lower order scheme like Upwind would not produce viable results due to numerical diffusion. Since the discretisation schemes being used in this investigation are Finite Volume Methods (approximations), a model that provides a good wall-function approximation is desirable because the near-wall viscous layer will not be fully resolved by the simulation.

3.2 C_l and C_d results

LAMINAR FLOW at $Re = 275$ and $10°$ incidence					
Case File	**Grid**	**Upwind**		**QUICK**	
		C_l	C_d	C_l	C_d
airfoil-10.cas	(40x10)	6.925×10^{-1}	2.4013×10^{-1}	5.6635×10^{-1}	1.8579×10^{-1}
airfoil-20.cas	(80x20)	6.5178×10^{-1}	2.1338×10^{-1}	5.7775×10^{-1}	1.7726×10^{-1}
airfoil-40.cas	(160x40)	6.2727×10^{-1}	1.9473×10^{-1}	5.868×10^{-1}	1.7478×10^{-1}
airfoil-80.cas	(320x80)	6.126×10^{-1}	1.837×10^{-1}	5.906×10^{-1}	1.734×10^{-1}
TURBULENT FLOW at $Re = 1.2 \times 10^6$ and $0°$ incidence					
airfoil-turb.cas	(220x100)	XX	XX	1.5594×10^{-1}	1.4371×10^{-2}

Table 2: Predicted C_l and C_d values for different aerofoil cases.

Table 2 establishes that as grid density increases (for either convection scheme), C_l and C_d converge because the number of nodes doubles for each subsequent laminar case, yet the magnitude of the difference between the previous value decreases each time. For the 320×80 grid, the numerical discretisation errors have been reduced to a negligible level, so no significant changes are seen if the grid is refined further **(3)**.

The behaviour seen in this table reflects what one might expect from a first and third order discretisation scheme as the grid is refined because a finer mesh and higher order convection scheme will be able to more accurately compute near-surface gradients, therefore reducing errors for areas in which they are the largest.

The vector, contour and coefficient of pressure plots for laminar flow from Section 2 all provided evidence that C_l and C_d would be more accurate as the grid was refined however it was the residual plots that allowed one to identify the grid independent solution because Figure 5 directly compared computational error values. These errors decreased exponentially for the third order scheme but only linearly for the first order scheme, meaning the finest mesh for the QUICK scheme is the grid independent solution in this case.

4 Conclusion

This report highlighted the importance of selecting the correct mesh refinement and convection schemes in order to output the most accurately converged C_l and C_d values. The results from turbulent flow were compared to experimental data and it was established that the model used was accurate for this investigation, as it produced feasible results that reflected theoretical predictions. Overall, it was demonstrated that the QUICK (3rd order) scheme was more accurate than the Upwind (1st order), as it was able to reach the grid-independent solution for the finest grids, which is the behaviour one would expect.

An appropriate level of grid independence was achieved for both the laminar 320 $\times$ 80 and turbulent 220 $\times$ 100 meshes, which indicated that mesh refinement produced negligible changes in C_l and C_d values, therefore making these solutions independent of grid size.

5 References

(1) - Airfoil Laboratory Handout (2025), AERO30052

(2) - Nakayama, A., (1985), Characteristics of the flow around conventional and supercritical airfoils, J. Fluid Mech., 160, 155-179

(3) - Tim Craft (2025), Finite Difference Approximation and Modelling lecture notes, slide 8